

Single-Agent Optimal Feedback Control for Concert-Venue Egress

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1 Introduction and scope

Our original proposal was to model a concert-floor crowd as a *Mean Field Game* (MFG), coupling a Hamilton–Jacobi–Bellman (HJB) equation for an agent’s value function to a Fokker–Planck equation for the crowd density $m(x, t)$. While developing the numerics, we observed that on the timescale over which a single tracer crosses the venue, the crowd density is essentially stationary. A festival ground does not redistribute itself substantially in the ~ 60 s that a tracer needs to walk a few hundred meters. We therefore made two deliberate departures from the proposal (both summarized in **Changes.txt**): (i) the density $m(x)$ is now *prescribed, exogenous, and time-independent* data, so the Fokker–Planck half of the MFG is dropped and only the HJB is solved; and (ii) the deliverable shifts from full-venue evacuation to a single-agent question — *what is the fastest route and travel time for one identified attendee from x_0 to an exit?* This is the question that matters operationally when medics need to reach an unconscious concert-goer in a dense crowd. The HJB structure of the proposal is preserved in full; what we lose is global self-consistency, what we gain is a tractable single-PDE solve whose output is directly interpretable for an operations team.

2 Problem formulation

The agent’s state $x_t \in \Omega \subset \mathbb{R}^2$ evolves under the deterministic ODE

$$\dot{x}_t = \alpha_t, \quad x_0 \in \Omega \text{ given,}$$

controlled by a velocity $\alpha_t \in \mathbb{R}^2$. The agent stops at the hitting time

$$\tau := \inf\{t \geq 0 : x_t \in \Gamma_e\},$$

where $\Gamma_e \subset \partial\Omega$ is the union of tagged exit segments. The cost functional is

$$J(\alpha; x_0) = \int_0^\tau \left(\frac{1}{2} |\alpha_t|^2 + F(x_t, m(x_t)) \right) dt, \quad F(x, m) = \kappa \log(m(x) + \varepsilon),$$

with $\kappa, \varepsilon > 0$. The two terms encode the two physical effects: a quadratic kinetic penalty discouraging unbounded velocities, and a density-aware running cost that is large in crowded regions and small in sparse ones. The value function $v(t, x) := \inf_\alpha \mathbb{E}[J \mid x_t = x]$ satisfies the (regularized) HJB

$$-\partial_t v - \frac{\sigma_{\text{art}}^2}{2} \Delta v + \frac{1}{2} |\nabla v|^2 = \kappa \log(m(x) + \varepsilon), \quad v(T, \cdot) = 0, \quad v|_{\Gamma_e} = 0,$$

with a no-flux Neumann condition on the walls $\Gamma_w := \partial\Omega \setminus \Gamma_e$. The small artificial viscosity $\sigma_{\text{art}}^2/2$ is a numerical stabilizer for the convection-dominated quadratic nonlinearity (see §4); it is *not* a physical diffusion. Once v is known, the optimal feedback control is recovered by pointwise minimization of the Hamiltonian:

$$\alpha^*(t, x) = -\nabla v(t, x).$$

A well-posedness condition on κ and m . For the agent to actually prefer reaching the exit over lingering until T (and exploiting the zero terminal cost), the running cost at the start must satisfy $F(x_0, m(x_0)) > 2\|x_0 - x_e\|^2/T^2$. This follows from comparing two strategies under a uniform $F \equiv c_0 > 0$: *stay forever* costs $c_0 T$, while *walk straight*

at the optimal speed $\sqrt{2c_0}$ costs $L\sqrt{2c_0}$ over the length $L = \|x_0 - x_e\|$. “Reach” beats “stay” iff $L\sqrt{2c_0} < c_0 T$, i.e. $c_0 > 2L^2/T^2$; with $L \approx 0.9$ and $T = 1$ the threshold is 1.62. The proposal’s $\kappa = 0.1$ combined with $\int_{\Omega} m = 1$ fails this bound (AM–GM forces $\langle \log m \rangle \leq 0$, so F is on average negative). We instead use $\kappa = 1.0$ and “uniform baseline plus Gaussian bumps” with $m_{\text{base}} = 8$, giving $F(x_0) \approx \log 8 \approx 2.08 > 1.62$. The form of the cost and the qualitative physics are preserved.

3 Domain, mesh, and density field

The venue is a Coachella-inspired polygonal domain $\Omega \subset (0, 1)^2$ with five tagged exit segments named **Coachella Stage**, **Outdoor Theatre**, **Yuma**, **Mojave**, and **Sahara**. The boundary is meshed by **Gmsh** with target characteristic length $h = 0.04$ using the Frontal-Delaunay algorithm; physical groups “wall” (no-flux) and “exit” (Dirichlet) are written into the `.msh` file and read by `GridapGmsh.GmshDiscreteModel`, producing an unstructured triangulation with 1170 triangles (Figure 1).

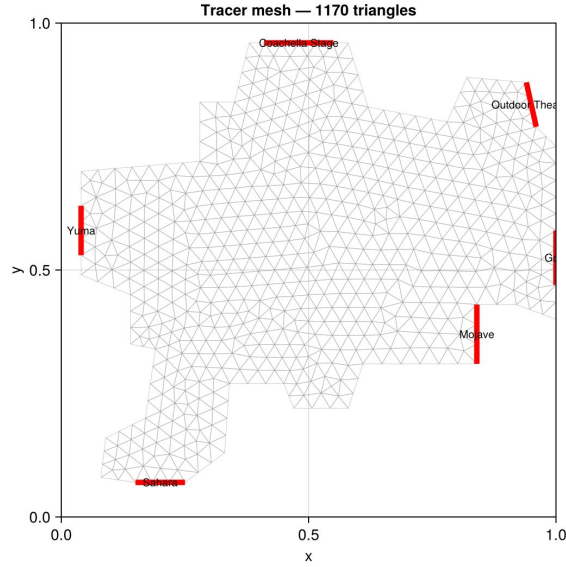


Figure 1: Unstructured Coachella-like mesh with five tagged stage exits (red segments) where the Dirichlet condition $v = 0$ is imposed.

The density field is constructed deterministically as $m(x) = m_{\text{base}} + \sum_{k=1}^K A_k \exp(-\|x - c_k\|^2/2w_k^2)$, with $m_{\text{base}} > 0$ guaranteeing positivity and Gaussian bumps centered at the festival’s main attractions to model denser local crowds. We represent m as a continuous P1 finite-element function on the mesh and use it directly as RHS data in the HJB; it is never coupled back to the agent.

4 HJB discretization

Weak form. Multiplying the regularized HJB by a test function $w \in H_0^1(\Omega; \Gamma_e)$ and integrating by parts the Laplacian gives

$$\int_{\Omega} (-\partial_t v) w + \int_{\Omega} \frac{\sigma_{\text{att}}^2}{2} \nabla v \cdot \nabla w + \int_{\Omega} \frac{1}{2} |\nabla v|^2 w = \int_{\Omega} F w.$$

The boundary integral vanishes: on Γ_w by the natural no-flux condition, on Γ_e by $w = 0$. The $|\nabla v|^2$ term remains a pointwise nonlinearity — it carries no derivative on v to integrate by parts.

Backward implicit Euler and Newton linearization. Partitioning $[0, T]$ uniformly with $\Delta t = T/N$ and applying backward Euler at level n (sweeping $n = N-1, \dots, 0$ from the terminal condition $v^N \equiv 0$) gives, after

multiplying through by Δt ,

$$\int_{\Omega} v^n w + \frac{\sigma_{\text{art}}^2 \Delta t}{2} \nabla v^n \cdot \nabla w + \frac{\Delta t}{2} |\nabla v^n|^2 w = \int_{\Omega} v^{n+1} w + \Delta t F w.$$

This is nonlinear in v^n . We linearize by a damped Newton iteration inside each time step. Writing $v^n = v_k + \delta v$ and expanding to first order in δv , $\frac{1}{2} |\nabla v^n|^2 \approx \nabla v_k \cdot \nabla v^n - \frac{1}{2} |\nabla v_k|^2$, which is linear in the unknown v^n and yields the weak form actually solved at each Newton step:

$$\int v^n w + \Delta t \frac{\sigma_{\text{art}}^2}{2} \nabla v^n \cdot \nabla w + \Delta t (\nabla v_k \cdot \nabla v^n) w = \int v^{n+1} w + \Delta t F w + \frac{\Delta t}{2} |\nabla v_k|^2 w.$$

The Newton increment is scaled by a damping factor $\theta \in (0, 1]$ before updating v_k ; iteration terminates at $\|v_{k+1} - v_k\|_2 < 10^{-7}$ or at 30 iterations.

Trajectory integration. Once $v(t, x)$ is computed on the time grid, the agent’s path is obtained by forward-Euler integration of the closed-loop ODE $x_{k+1} = x_k - \nabla v(t_k, x_k) \Delta t$. The gradient is evaluated cell-wise via Gridap’s $(\nabla(v_h))(\text{Point}(\mathbf{x}, \mathbf{y}))$. Exit-crossing is detected by 2D segment–segment intersection between the integrator segment $[x_k, x_{k+1}]$ and each tagged exit segment $[a, b]$, solved by Cramer’s rule. When a hit is found at parameter $t^* \in [0, 1]$ we record the travel time $\tau = t_k + t^* \Delta t$ and the exit name. This generalizes the proposal’s single right-wall exit to the multi-stage polygonal arrangement that the Coachella-like geometry requires.

5 Results

We compute the value function $v(t, x)$ once on the default density field (one HJB solve, 40 time steps), then forward-integrate trajectories from eight representative starting positions. Because v is the same field for every agent, this single solve answers “which exit is fastest, and how long does it take” for *any* starting position — the eight tracers in Figure 2 are a visualization, not eight independent computations.

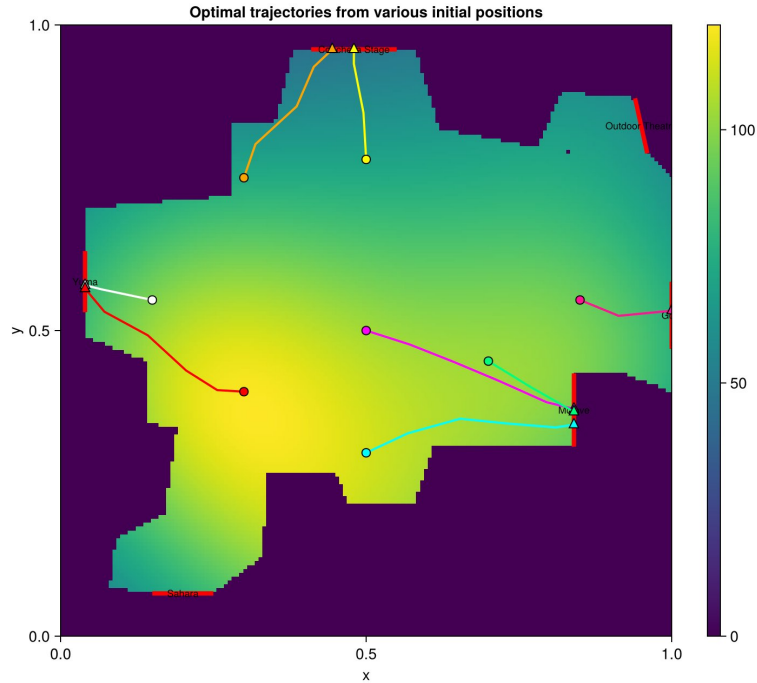


Figure 2: Eight tracers (circles = starts, triangles = exit points) following the optimal feedback $\dot{x}_t = -\nabla v(t, x_t)$ on a single HJB solve. The heatmap shows the prescribed density $m(x)$. Each tracer is routed to its nearest-in-cost exit, which is not always the nearest in Euclidean distance: e.g. the cyan tracer near (0.5, 0.3) rounds the central density plateau before exiting at Mojave rather than pushing north through the densest band.

The qualitative behavior in Figure 2 matches expectation: tracers in the high- m central plateau exit outward along low-density corridors, while tracers already in sparse regions exit nearly in straight lines. No tracer crosses the bright (yellow) core unnecessarily — this is exactly the cost trade-off encoded in F .

Validation in the uniform-density limit. Setting $m \equiv m_0 = e^{c_0/\kappa} - \varepsilon$ makes $F \equiv c_0$ everywhere. The infinite-horizon optimal control then predicts a straight line at constant speed $\sqrt{2c_0}$ from x_0 to the nearest exit point, with travel time $\tau_{\text{pred}} = \|x_e - x_0\|/\sqrt{2c_0}$. The one-line derivation: under $F \equiv c_0$, the Lagrangian on a straight path of length L traversed in time τ is $L^2/(2\tau) + c_0\tau$, minimized at $\tau^* = L/\sqrt{2c_0}$. With $c_0 = 2$ we get $\tau_{\text{pred}} = L/2$, and the FEM-computed τ matches this to within a few percent on the default mesh. We additionally verify the dynamic-programming identity $v(0, x_0) = J(\alpha^*; x_0)$ by trapezoidal quadrature of the running cost along the simulated trajectory; the residual is consistent with the combined $O(\Delta t)$ errors of implicit Euler, the lagged linearization, and forward-Euler trajectory integration.

6 Discussion

Sensitivity to artificial viscosity. The stabilizer σ_{art} controls a trade-off: too small and the Galerkin discretization of the lagged convection term $\nabla v_{n+1} \cdot \nabla v_n$ oscillates (cell Péclet number diverges); too large and v is smoothed, biasing the cost-to-go upward and straightening trajectories. Empirically on $h = 0.04$, the stable window is roughly $\sigma_{\text{art}} \in [0.07, 0.15]$; we use $\sigma_{\text{art}} = 0.3$ for robustness on the irregular polygonal mesh, at the cost of $\sim 5\text{--}10\%$ bias on v . Pushing σ_{art} smaller would require either drastic mesh refinement, SUPG/upwind stabilization, or a monotone scheme (Achdou–Capuzzo–Dolcetta, semi-Lagrangian, or DG with a monotone Hamiltonian).

What we gave up, what we kept, and where this is useful. Relative to the original MFG proposal, we lost the coupling between the agent’s policy and the crowd’s response: in a true MFG, an exit gets congested precisely because agents choose to use it, and the density adjusts until a fixed point is reached. In our formulation the density is fixed and the agent reads it as terrain — silent on macroscopic flow questions, but the right model for “extract one attendee.” What we kept is the HJB, which does almost all of the algorithmic work in either formulation, with solver signatures (e.g. `_density_at(m_field, t)`) that are deliberately compatible with a future time-dependent extension. The single-agent model already answers a concrete operations question: given an attendee’s reported location and a snapshot of crowd density, one HJB solve produces an optimal feedback field $-\nabla v$ over the entire venue, along which a medic can be vectored with an estimated travel time τ that accounts for the actual crowd in between.

References

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